

Jawaharlal Nehru Govt. Engineering College Sundernagar, Mandi (H.P.)

Course Name: **Applied Physics** Course Code: PHY-111 Examination: 1st Periodical

Scheme: NEP/CBCS

Branch: CE/ECE/CSE

Semester: 1st

Faculty Name: **Dr. Neelam Guleria**

Question Paper for 1st Periodical Examinations (September/October, 2024)

Time: 2 Hours

Max. Marks: 50

Note: Attempt all questions.

Q. No.	Sub Q. No.	CO(s)	Question	Maximum Marks for Q/Sub Q.
Q1.	(i)		State forced oscillations.	1
	(ii)		What will be the effect on time period of simple pendulum if its mass is doubled?	1
	(iii)	CO3	What is meant by term sharpness of resonance?	1
	(iv)		Define logarithmic decrement.	1
	(v)		Calculate the maximum velocity of particles that executes SHM of amplitude 0.06 m with time period of 10π sec.	1
	(vi)		What does LASER stands for?	1
	(vii)		What is the significance of metastable state?	1
	(viii)	CO4	Find the ratio of population of the two states in a laser that produces a light of wavelength $6500 \text{ \AA}$ at 30°C .	1
	(ix)		The population of the various energy levels of a system in thermal equilibrium are given by: a) Boltzmann distribution law b) Einstein relation c) Planck's Law d) Beer's Law	1
	(x)		If He-Ne laser emits laser light at $1.15 \mu\text{m}$, find the energy difference between the two lasing levels in eV.	1
Q2.		CO3	Derive a differential equation of motion for damped harmonic motion and discuss the case of Over-damped & Under-damped motion.	10
Q3.		CO3	Define Simple Harmonic motion and derive expression for total mechanical energy of a particle executing SHM.	10
Q4.		CO1, CO4	Define stimulated absorption, stimulated emission and spontaneous emission. Derive the relation between Einstein's coefficients?	10
Q5.		CO1, CO4	Explain the principle, construction, working & output of Ruby Laser. Or Explain the principle, construction, working & output of He-Ne Laser.	10

Course Outcomes (CO): Upon successful completion of the course, students will be able to:

CO1. Recognize the importance of Physics in describing different physical phenomenon

CO2. Comprehend the Special theory of relativity by examining inertial & non-inertial frames, Michelson-Morley experiment, Galilean & Lorentz transformations.

CO3. Demonstrate the concept of harmonic oscillators by studying simple harmonic, damped harmonic and forced harmonic oscillations

CO4. Illustrate the concept of LASER, Optical Fibres and X-rays in various fields.

CO5. Appraise the concept of electromagnetic waves and Superconductivity in the field of Engineering

CO6. Recognize the inadequacy of classical mechanics for certain physical problems and thus find the solution of these problems using Quantum Mechanics.